

AN AI ASSISTANT ON YOUR OWN MACHINE

Meru runs open-weight models on a machine you control. You need no API key and pay for no tokens, so a question costs electricity, and the model on your disk today still runs a year from now. No prompt leaves the machine unless you allow it.



github.com/aarora79/meru
Architecture in three levels:
aarora79.github.io/meru/

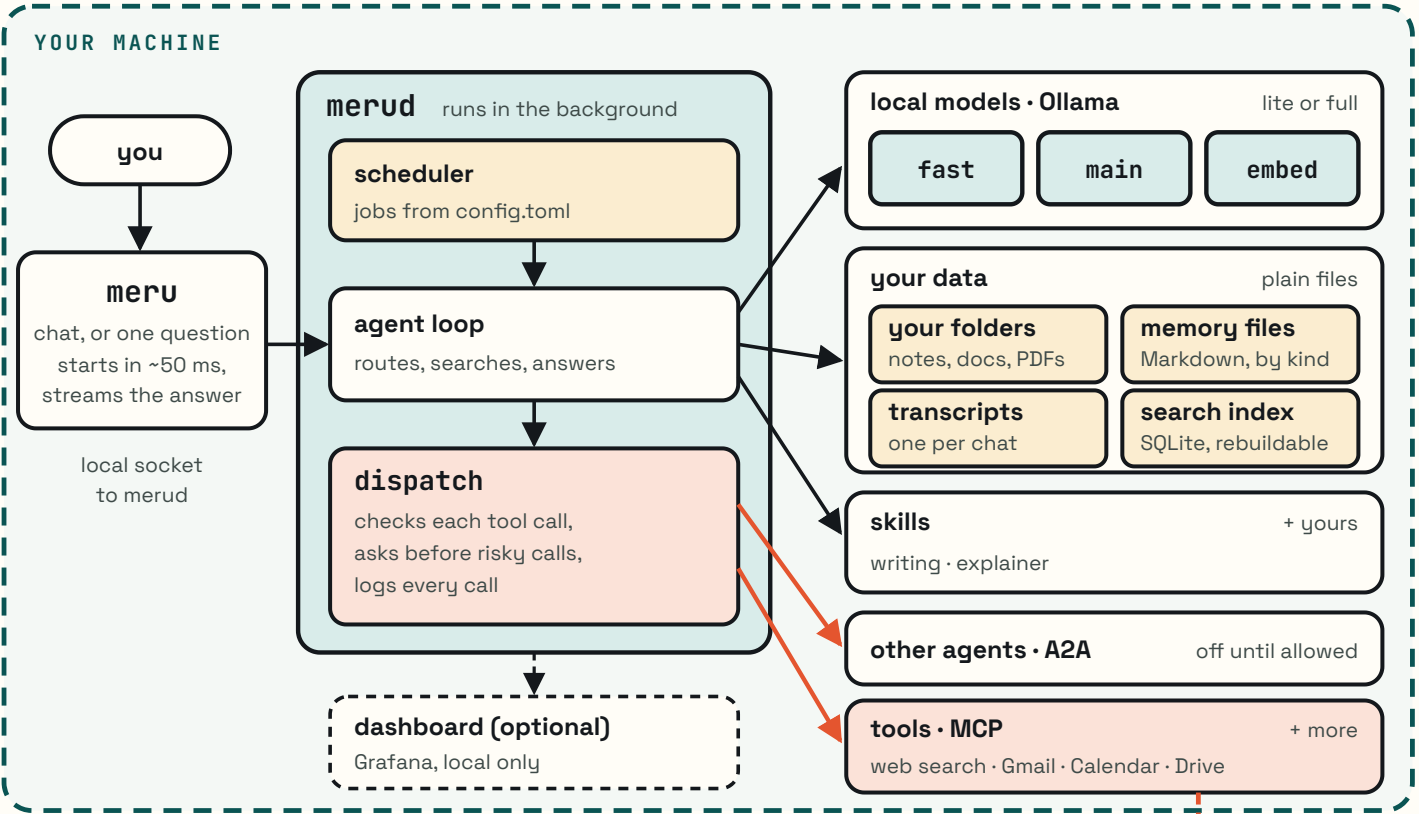


मेरु

Meru is the cosmic mountain that the sun, moon and stars turn around. This assistant takes the name because it works the same way: it stays in one place, on your machine, and your notes, tools and daily routine turn around it.

01 EVERYTHING RUNS INSIDE THE DASHED LINE

the parts of Meru



Inside the line: the models, your data and the assistant.

Black arrows stay on the machine. Coral arrows leave merud through dispatch, and only the tools you allow cross the line.

02 PICK THE PROFILE YOUR MACHINE CAN HOLD

lite is the default

TIER	JOB	LITE (DEFAULT)	FULL
fast	routing, query rewrites	MiniCPM5-2B, 1.6GB	same
main	reasoning, choosing tools	same model as fast	qwen3.8:27b, ~18GB at 4-bit
embed	vectors for search	nomic-embed-text, 768 dims	qwen3-embedding:0.6b, 1024 dims
needs	hardware	16GB of RAM; a CPU is enough	Apple silicon with 32GB+, or a ~24GB GPU

Platforms: macOS on Apple silicon (tested), Linux, including cloud VMs such as EC2, and Windows 10+ (untested at first).

03 THE RULES THE CODE KEEPS

from the privacy boundary

- Local models only.** The one model client talks to Ollama on this machine and refuses any other URL.
- Deny by default.** A new MCP server or A2A agent gives the model nothing until you allow tools by name.
- Every external action logged.** Review them all with `meru log`.
- No telemetry leaves the machine.** Meru measures itself locally, sends no crash reports and never checks for updates.
- Loopback unless you say so.** A remote server or agent needs `network = true` in its entry.
- No sandbox.** MCP servers run with your permissions, so add only ones you trust.

04 ONE TURN, FIVE STEPS, IN A LOOP WE WROTE

no agent framework · traced locally

1 · Route

fast rewrites the question, picks a route.

2 · Context

Skills, memories, hits, history, each on a budget.

3 · Main

main streams the answer to you, or asks for a tool.

4 · Dispatch

Checks, asks if needed, runs and logs each call.

5 · Repeat

Until the answer needs no tool, or 8 rounds.

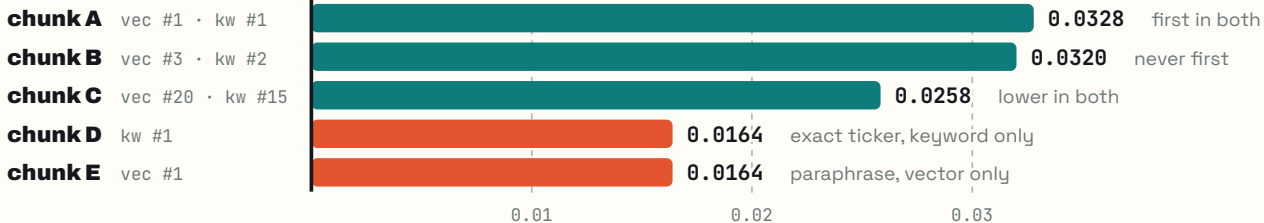
merud traces each step with OpenTelemetry and sends the data only to a loopback endpoint, such as the bundled Grafana, with no prompt text unless you opt in.

05 TWO SEARCHES, MERGED BY RANK

BM25 + sqlite-vec + reciprocal-rank fusion

Each turn searches your files, your memories and past chat summaries. Keyword search (BM25) finds exact strings such as tickers, and meaning search finds paraphrase. Go merges the two ranked lists.

$RRF\ SCORE = \sum \frac{1}{(60 + RANK)} \cdot RANKS\ INVENTED, SCORES\ COMPUTED$



06 EVERY TOOL CALL PASSES ONE FUNCTION

dispatch · deny by default



07 MEMORY IS A FOLDER OF MARKDOWN FILES

~/meru/memory/ · one fact per file

me/ who you are
preferences/ how you like things done
projects/ work, goals, deadlines
people/ people you mention
reference/ where things live
other/ anything else

- Each file carries its date and source chat. Delete the file and Meru forgets it.
- The model saves one with the built-in remember tool, through dispatch.
- Each chat ends with a short summary, so “what did we decide last week?” finds it.
- Files hold the truth. The SQLite index, meru.db, rebuilds from them.

08 SETUP TAKES ONE COMMAND

meru setup · skills · jobs

MERU SETUP · FIRST RUN

- 1 Checks for Ollama and prints the install command if needed.
- 2 Downloads the lite or full models, your pick.
- 3 Writes config.toml and asks which folders to index.
- 4 Offers starter MCP servers (web search, Gmail, Drive). It adds each for you, or shows you what to paste.
- 5 Answers a test question.

SKILLS AND JOBS

- writing and explainer ship built in; each loads only when needed.
- Skills are Markdown in ~/meru/skills/. Your edits win.

```
[[jobs]]
name = "morning-brief"
schedule = "0 7 * * 1-5" # weekdays 7:00
prompt = "Brief me on today..."
```

09 SHIPPING ORDER

each milestone useful by itself

v0.1

IT TALKS

meru "hello" shows a first token in under 1s.

v0.2

IT KNOWS YOUR FILES

It answers from a local note and cites the file.

v0.3

IT CAN DO THINGS

It answers by calling an MCP server you already run.

v0.4

IT KNOWS YOU

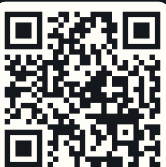
It recalls something from last week without a reminder.

v0.5

IT ACTS UNPROMPTED

A morning brief arrives without you asking.

Pre-alpha, design phase. The code will follow ARCHITECTURE.md. The 50ms start and first-token target are unmeasured goals; panel 05's ranks are made up.



READ THE DESIGN, FOLLOW THE BUILD

Code: github.com/aarora79/meru — README.md, ARCHITECTURE.md, ROADMAP.md

Architecture in three levels: aarora79.github.io/meru/

Amit Arora · MIT license · Ollama on loopback · Go binaries you can copy to another machine and run.